

SQL Queries

Users Table

```
CREATE TABLE dbo.Users
(
    UserId INT IDENTITY(1,1) NOT NULL,
    FullName NVARCHAR(100) NOT NULL,
    Email NVARCHAR(150) NOT NULL,
    Username NVARCHAR(50) NOT NULL,
    PasswordHash NVARCHAR(64) NOT NULL,
    Role NVARCHAR(20) NOT NULL,
    Phone NVARCHAR(30) NOT NULL,
    Address NVARCHAR(250) NULL,
    Status NVARCHAR(20) NOT NULL CONSTRAINT DF_Users_Status DEFAULT N'Active',
    CreatedAt DATETIME2(0) NOT NULL CONSTRAINT DF_Users_CreatedAt DEFAULT SYSDATETIME(),

    CONSTRAINT PK_Users PRIMARY KEY CLUSTERED (UserId),
    CONSTRAINT UQ_Users_Email UNIQUE (Email),
    CONSTRAINT UQ_Users_Username UNIQUE (Username),
    CONSTRAINT CK_Users_Role CHECK (Role IN (N'Customer', N'Owner', N'Admin', N'SuperAdmin')),
    CONSTRAINT CK_Users_Status CHECK (Status IN (N'Active', N'Suspended'))
);
```

Cars Table

```
CREATE TABLE dbo.Cars
(
    CarId INT IDENTITY(1,1) NOT NULL,
    OwnerId INT NOT NULL,
    CarName NVARCHAR(100) NOT NULL,
    Brand NVARCHAR(60) NOT NULL,
    Model NVARCHAR(60) NOT NULL,
    CarNumber NVARCHAR(50) NOT NULL,
    Seats INT NOT NULL,
    PricePerDay DECIMAL(12,2) NOT NULL,
    Location NVARCHAR(100) NOT NULL,
    Status NVARCHAR(20) NOT NULL CONSTRAINT DF_Cars_Status DEFAULT N'Available',
    Description NVARCHAR(500) NULL,
    CreatedAt DATETIME2(0) NOT NULL CONSTRAINT DF_Cars_CreatedAt DEFAULT SYSDATETIME(),

    CONSTRAINT PK_Cars PRIMARY KEY CLUSTERED (CarId),
    CONSTRAINT UQ_Cars_CarNumber UNIQUE (CarNumber),
    CONSTRAINT FK_Cars_Users_Owner FOREIGN KEY (OwnerId) REFERENCES dbo.Users(UserId),
    CONSTRAINT CK_Cars_Seats CHECK (Seats > 0),
    CONSTRAINT CK_Cars_PricePerDay CHECK (PricePerDay > 0),
    CONSTRAINT CK_Cars_Status CHECK (Status IN (N'Available', N'Rented', N'Maintenance', N'Unavailable'))
);
```

Bookings Table

```
CREATE TABLE dbo.Bookings
(
    BookingId INT IDENTITY(1,1) NOT NULL,
    CustomerId INT NOT NULL,
    CarId INT NOT NULL,
    RentDate DATE NOT NULL,
    ExpectedReturnDate DATE NOT NULL,
    ActualReturnDate DATE NULL,
    TotalAmount DECIMAL(12,2) NOT NULL,
    DiscountAmount DECIMAL(12,2) NOT NULL CONSTRAINT DF_Bookings_DiscountAmount DEFAULT 0,
    PayableAmount DECIMAL(12,2) NOT NULL,
    PaymentStatus NVARCHAR(20) NOT NULL CONSTRAINT DF_Bookings_PaymentStatus DEFAULT N'Unpaid',
    BookingStatus NVARCHAR(20) NOT NULL CONSTRAINT DF_Bookings_BookingStatus DEFAULT N'Active',
);
```

```

CreatedAt DATETIME2(0) NOT NULL CONSTRAINT DF_Bookings_CreatedAt DEFAULT SYSDATETIME(),

CONSTRAINT PK_Bookings PRIMARY KEY CLUSTERED (BookingId),
CONSTRAINT FK_Bookings_Users_Customer FOREIGN KEY (CustomerId) REFERENCES dbo.Users(UserId),
CONSTRAINT FK_Bookings_Cars FOREIGN KEY (CarId) REFERENCES dbo.Cars(CarId),
CONSTRAINT CK_Bookings_Dates CHECK (ExpectedReturnDate > RentDate),
CONSTRAINT CK_Bookings_ActualReturnDate CHECK (ActualReturnDate IS NULL OR ActualReturnDate >=
RentDate),
CONSTRAINT CK_Bookings_Amounts CHECK (TotalAmount >= 0 AND DiscountAmount >= 0 AND PayableAmount
>= 0),
CONSTRAINT CK_Bookings_PaymentStatus CHECK (PaymentStatus IN (N'Unpaid', N'Paid')),
CONSTRAINT CK_Bookings_BookingStatus CHECK (BookingStatus IN (N'Active', N'Completed', N'Cancelled'))
);

```

Offers Table

```

CREATE TABLE dbo.Offers
(
    OfferId INT IDENTITY(1,1) NOT NULL,
    OfferTitle NVARCHAR(120) NOT NULL,
    Description NVARCHAR(500) NULL,
    DiscountPercent DECIMAL(5,2) NOT NULL,
    StartDate DATE NOT NULL,
    EndDate DATE NOT NULL,
    OfferScope NVARCHAR(20) NOT NULL,
    CarId INT NULL,
    CreatedByUserId INT NOT NULL,
    Status NVARCHAR(20) NOT NULL CONSTRAINT DF_Offers_Status DEFAULT N'Active',
    CreatedAt DATETIME2(0) NOT NULL CONSTRAINT DF_Offers_CreatedAt DEFAULT SYSDATETIME(),

    CONSTRAINT PK_Offers PRIMARY KEY CLUSTERED (OfferId),
    CONSTRAINT FK_Offers_Cars FOREIGN KEY (CarId) REFERENCES dbo.Cars(CarId),
    CONSTRAINT FK_Offers_Users_CreatedBy FOREIGN KEY (CreatedByUserId) REFERENCES dbo.Users(UserId),
    CONSTRAINT CK_Offers_Discount CHECK (DiscountPercent > 0 AND DiscountPercent <= 100),
    CONSTRAINT CK_Offers_Dates CHECK (EndDate >= StartDate),
    CONSTRAINT CK_Offers_Scope CHECK (OfferScope IN (N'Platform', N'Owner', N'Car')),
    CONSTRAINT CK_Offers_Status CHECK (Status IN (N'Active', N'Inactive', N'Expired'))
);

```

Payment Table

```

CREATE TABLE dbo.Payments
(
    PaymentId INT IDENTITY(1,1) NOT NULL,
    BookingId INT NOT NULL,
    CustomerId INT NOT NULL,
    Amount DECIMAL(12,2) NOT NULL,
    PaymentMethod NVARCHAR(30) NOT NULL,
    TransactionNumber NVARCHAR(100) NOT NULL,
    PaymentStatus NVARCHAR(20) NOT NULL CONSTRAINT DF_Payments_PaymentStatus DEFAULT N'Paid',
    PaymentDate DATETIME2(0) NOT NULL CONSTRAINT DF_Payments_PaymentDate DEFAULT SYSDATETIME(),

    CONSTRAINT PK_Payments PRIMARY KEY CLUSTERED (PaymentId),
    CONSTRAINT UQ_Payments_TransactionNumber UNIQUE (TransactionNumber),
    CONSTRAINT FK_Payments_Bookings FOREIGN KEY (BookingId) REFERENCES dbo.Bookings(BookingId),
    CONSTRAINT FK_Payments_Users_Customer FOREIGN KEY (CustomerId) REFERENCES dbo.Users(UserId),
    CONSTRAINT CK_Payments_Amount CHECK (Amount > 0),
    CONSTRAINT CK_Payments_Method CHECK (PaymentMethod IN (N'bKash', N'Nagad', N'Rocket', N'Card', N'Cash')),
    CONSTRAINT CK_Payments_Status CHECK (PaymentStatus IN (N'Paid', N'Failed', N'Refunded'))
);

```

Fines Table

```
CREATE TABLE dbo.Fines
(
    FineId INT IDENTITY(1,1) NOT NULL,
    BookingId INT NOT NULL,
    LateDays INT NOT NULL,
    FineAmount DECIMAL(12,2) NOT NULL,
    FineStatus NVARCHAR(20) NOT NULL CONSTRAINT DF_Fines_FineStatus DEFAULT N'Unpaid',
    CreatedAt DATETIME2(0) NOT NULL CONSTRAINT DF_Fines_CreatedAt DEFAULT SYSDATETIME(),

    CONSTRAINT PK_Fines PRIMARY KEY CLUSTERED (FineId),
    CONSTRAINT UQ_Fines_BookingId UNIQUE (BookingId),
    CONSTRAINT FK_Fines_Bookings FOREIGN KEY (BookingId) REFERENCES dbo.Bookings(BookingId),
    CONSTRAINT CK_Fines_LateDays CHECK (LateDays >= 0),
    CONSTRAINT CK_Fines_FineAmount CHECK (FineAmount >= 0),
    CONSTRAINT CK_Fines_Status CHECK (FineStatus IN (N'Unpaid', N'Paid'))
);
```

Reviews Table

```
CREATE TABLE dbo.Reviews
(
    ReviewId INT IDENTITY(1,1) NOT NULL,
    BookingId INT NOT NULL,
    CustomerId INT NOT NULL,
    CarId INT NOT NULL,
    OwnerId INT NOT NULL,
    Rating INT NOT NULL,
    Comment NVARCHAR(1000) NOT NULL,
    CreatedAt DATETIME2(0) NOT NULL CONSTRAINT DF_Reviews_CreatedAt DEFAULT SYSDATETIME(),

    CONSTRAINT PK_Reviews PRIMARY KEY CLUSTERED (ReviewId),
    CONSTRAINT UQ_Reviews_BookingId UNIQUE (BookingId),
    CONSTRAINT FK_Reviews_Bookings FOREIGN KEY (BookingId) REFERENCES dbo.Bookings(BookingId),
    CONSTRAINT FK_Reviews_Users_Customer FOREIGN KEY (CustomerId) REFERENCES dbo.Users(UserId),
    CONSTRAINT FK_Reviews_Cars FOREIGN KEY (CarId) REFERENCES dbo.Cars(CarId),
    CONSTRAINT FK_Reviews_Users_Owner FOREIGN KEY (OwnerId) REFERENCES dbo.Users(UserId),
    CONSTRAINT CK_Reviews_Rating CHECK (Rating BETWEEN 1 AND 5)
);
```